



Electronics E-Commerce Sales & Customer Behavior Analysis

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Jessica Agnesia Tataung

EDUCATION

Agust 2025 -
Present

Data Science Bootcamp
dibimbing.id

2020 -
2024

Bachelor of pharmacy
Institut Teknologi Sumatera



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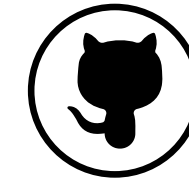
Previous project

These are several projects I explored during my learning journey with Dibimbing.id

01

Exploratory Data Analysis (EDA) on E-commerce sales

This section performs to understand the data structure, identify key patterns, and uncover insights related to customer behavior and sales performance.



[EDA E-commerce sales](#)

02

Advanced SQL Project on DVD Rental Database

This project is focusing on applying advanced SQL techniques in PostgreSQL. Students use subqueries, window functions, CTEs, and CASE WHEN statements.



[Advanced SQL Project](#)

03

Customer segmentation on E-commerce dataset

This project focuses on grouping customers based on behavioral and transactional patterns to support targeted marketing and customer retention strategies.



[Customer segmentation](#)

PROJECT BACKGROUND



Between September 2020 and February 2021:

- Revenue grew +26.79%
- Customer base declined by -27.72%
- Only 36% of customers make repeat purchases
- Majority of users drop off at the view stage

Revenue growth appears strong, but underlying customer behavior signals sustainability risk

BUSINESS OBJECTIVES & KEY QUESTIONS

Project Objective

- Identify growth drivers
- Evaluate retention
- Analyze conversion funnel

Key Questions

1. What drives revenue growth?
2. How strong is retention?
3. Where is the biggest drop-off?
4. Is revenue too concentrated?

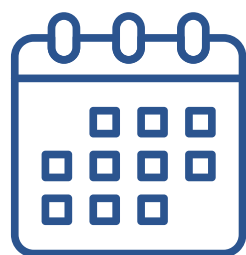
Business Hypotheses

- Growth driven by higher spending
- Retention drops after first purchase
- Revenue concentrated in top segments
- Funnel inefficiency limits conversion

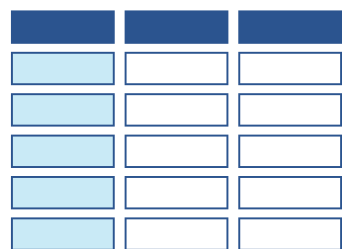
DATA UNDERSTANDING



E-Commerce events history in electronics store



2020-2021



event_time, event tpe, product_id, category_id,
category_code, brand, price, user_id, user_session



- 885K user interaction records
- 3 key behaviors: View → Cart → Purchase
- missing values in category code, brand & user session



[Link dataset E-Commerce events history.](#)

```
# Summary dataset  
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 885129 entries, 0 to 885128  
Data columns (total 9 columns):  
#      Column              Non-Null Count  Dtype  
---  -  
0     event_time            885129 non-null  object  
1     event_type            885129 non-null  object  
2     product_id            885129 non-null  int64  
3     category_id           885129 non-null  int64  
4     category_code         648910 non-null  object  
5     brand                 672765 non-null  object  
6     price                 885129 non-null  float64  
7     user_id               885129 non-null  int64  
8     user_session          884964 non-null  object  
dtypes: float64(1), int64(3), object(5)  
memory usage: 60.8+ MB
```


DATA PREPROCESSING

object → datetime
int64 → string
object → category

Covertng data type

Step 1

missing value → unknown

Handling missing value

Step 3

Exploratory Data Analysis

Step 5

Step 2

Removing duplicates

drop 656 duplicated

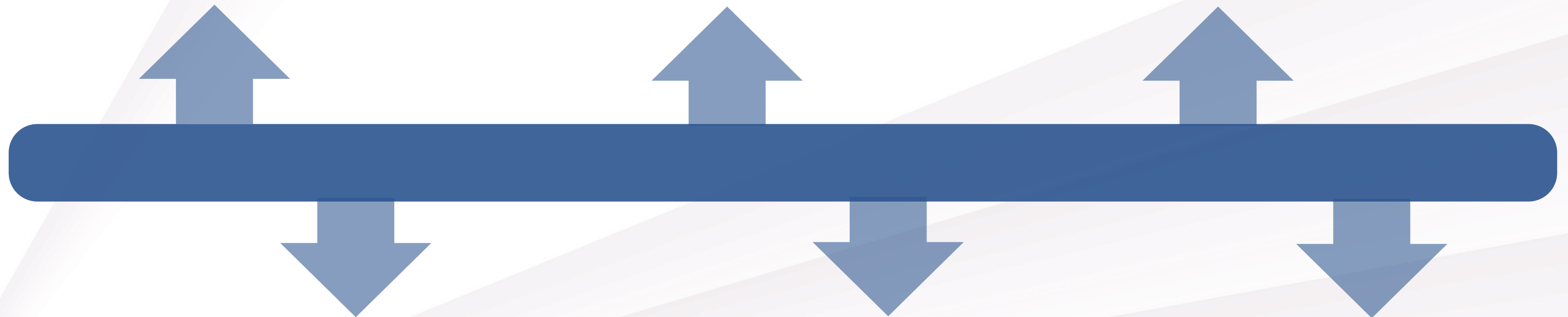
Step 4

make feature engenering

Step 6

Save preprocessed data

Save data as csv



Business Performance Snapshot (Sep 2020 – Feb 2021)

4.55 M

Revenue

+26.79%

Growth

21,304

Customer

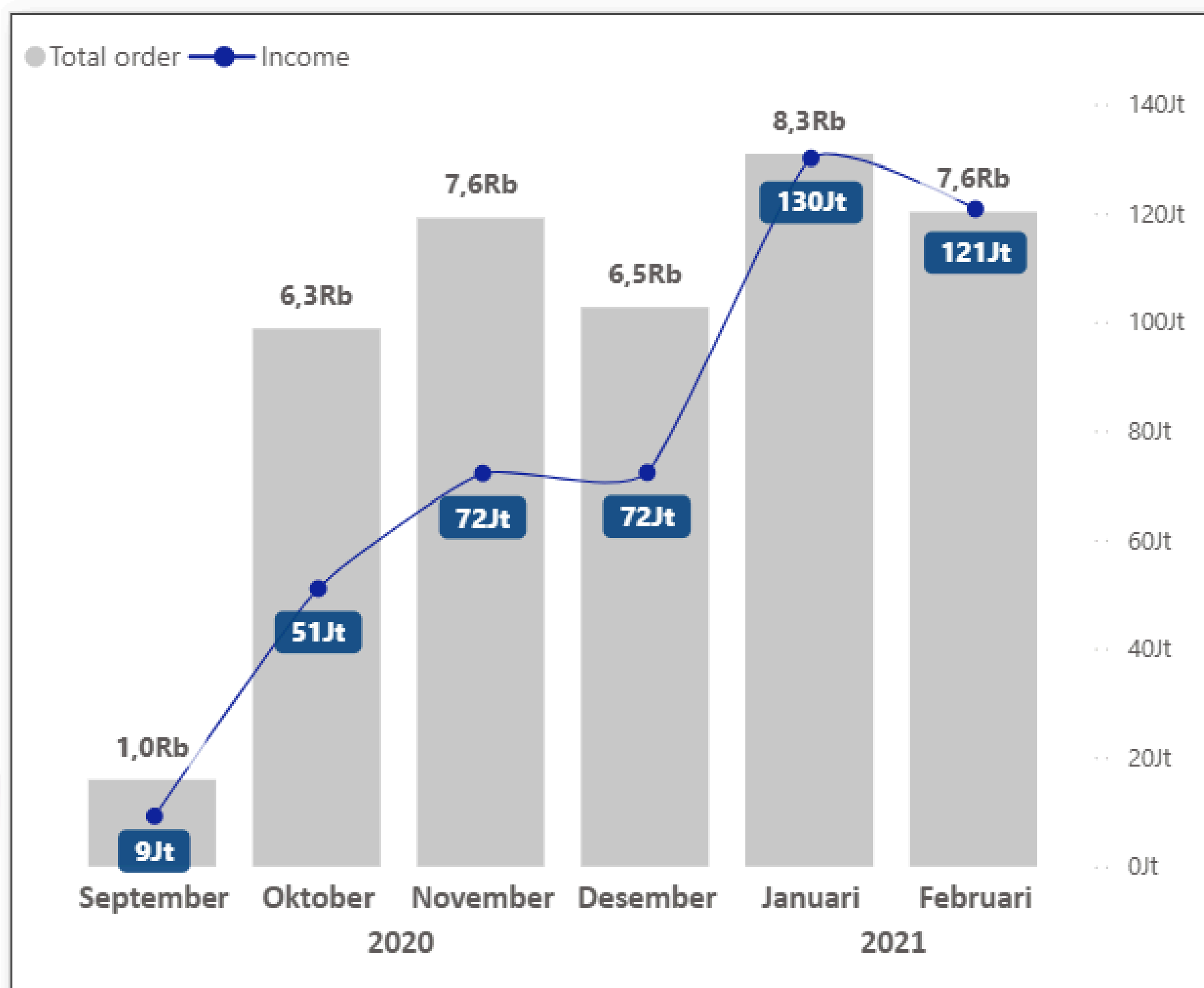
37,343

Orders

36%

Repeat rate

Sales & Revenue Metrics Overview (Sep 2020 - Feb 2021)



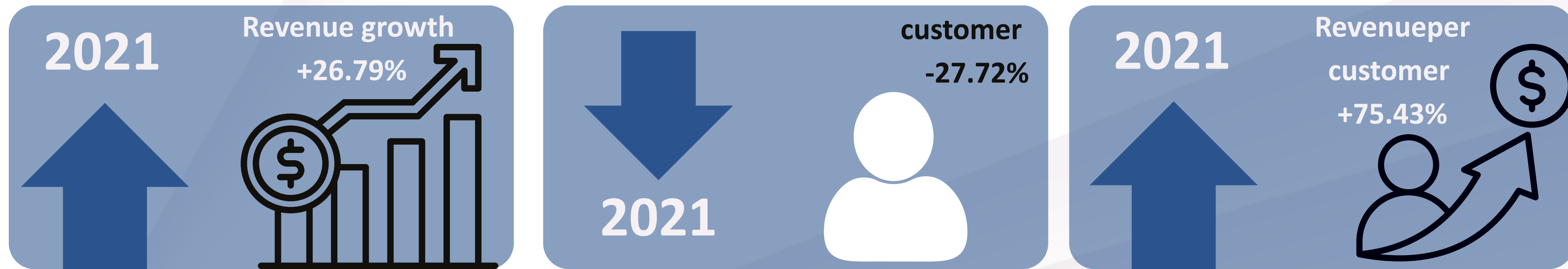
Revenue increased steadily from September and peaked in January 2021

In 2021, revenue grew by +26.79%, indicating strong early year momentum



Revenue Growth Driven by Higher Customer Spending

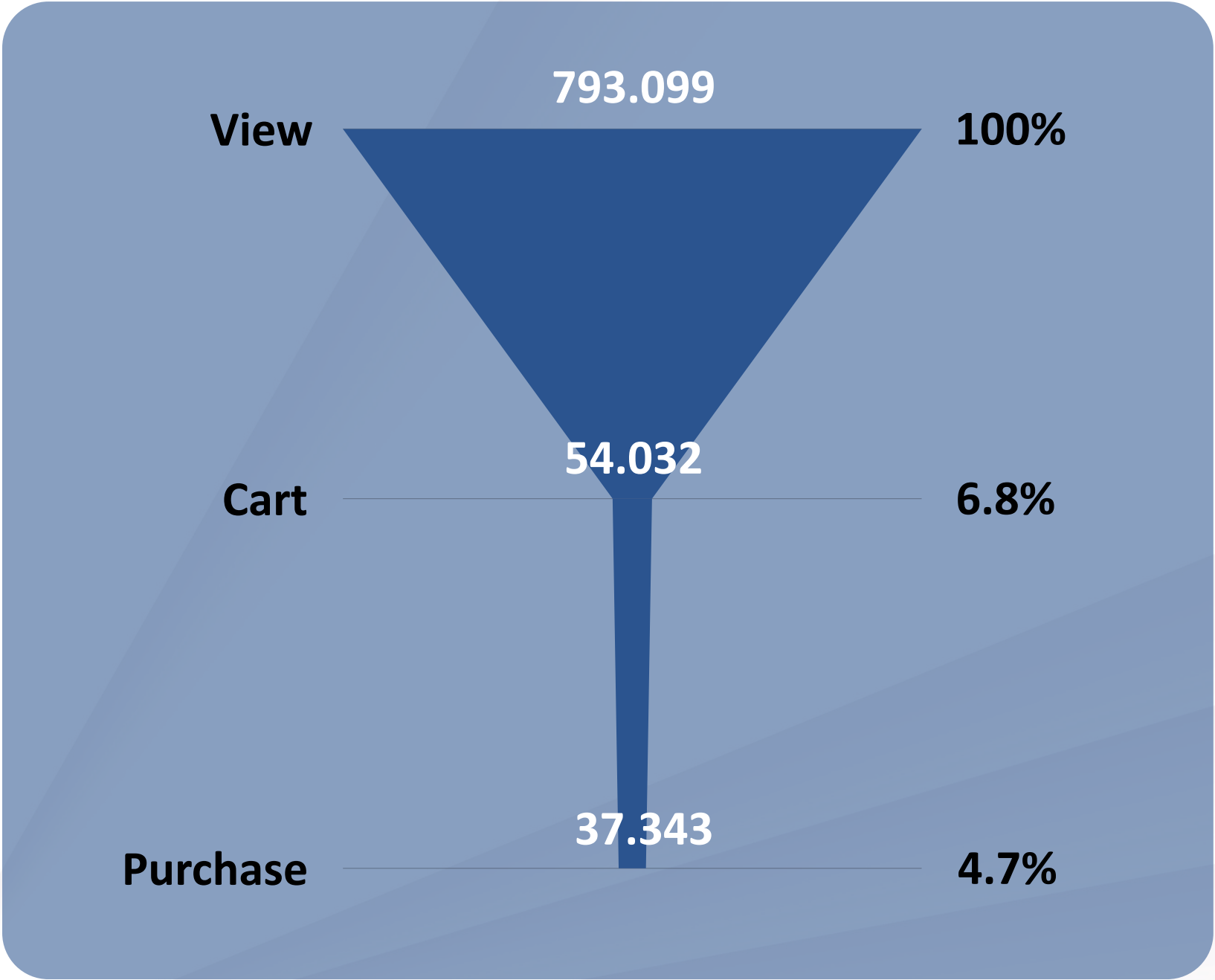
Despite Declining Customer Base



Revenue grew by 26.79% in 2021, despite a 27.72% decline in total customers

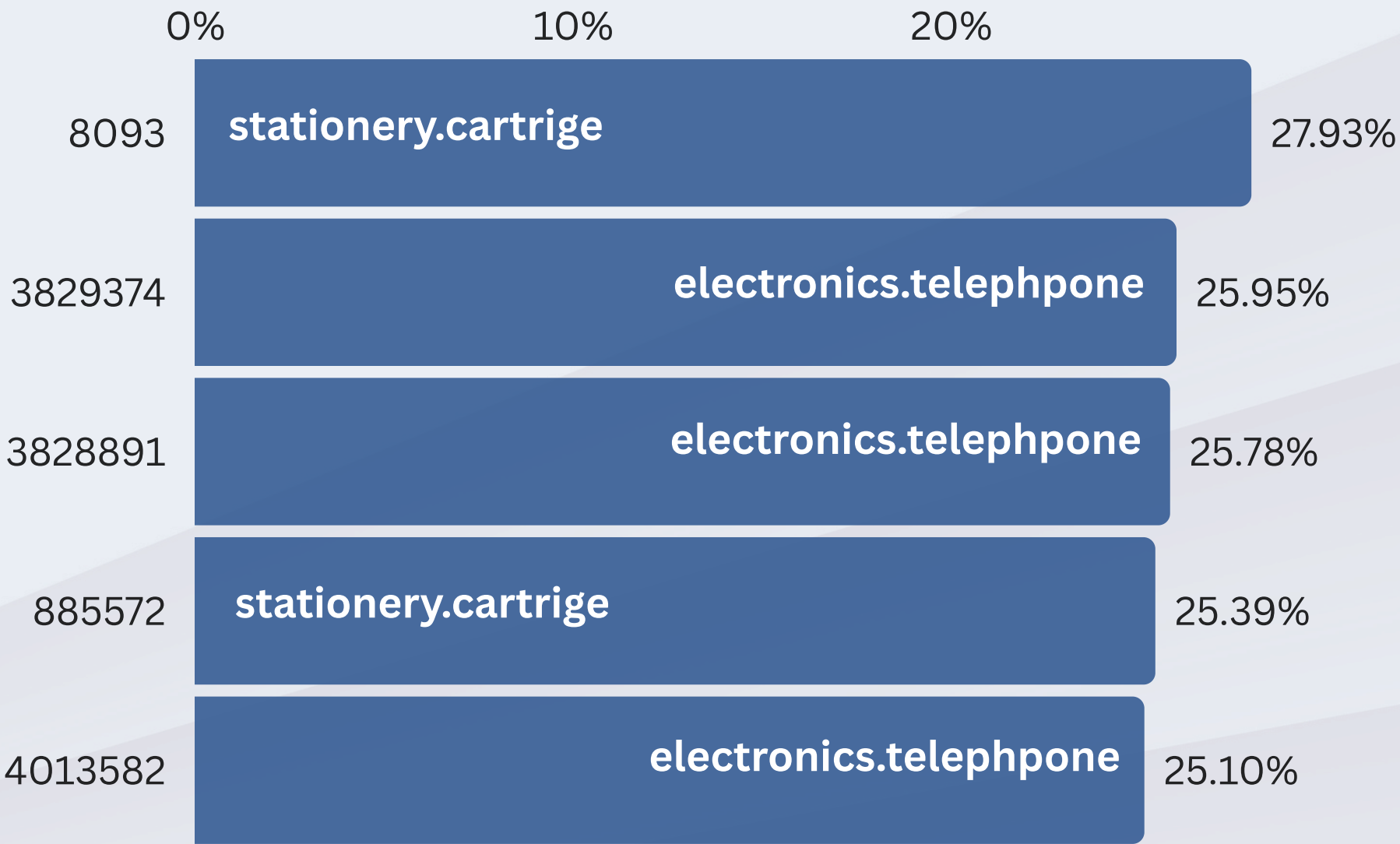
This growth was driven by a 75.43% increase in revenue per customer, indicating higher customer spending rather than customer acquisition

Only 4.7% of Views Convert into Purchases



Top 5 Product IDs

The Highest View to Purchase Conversion

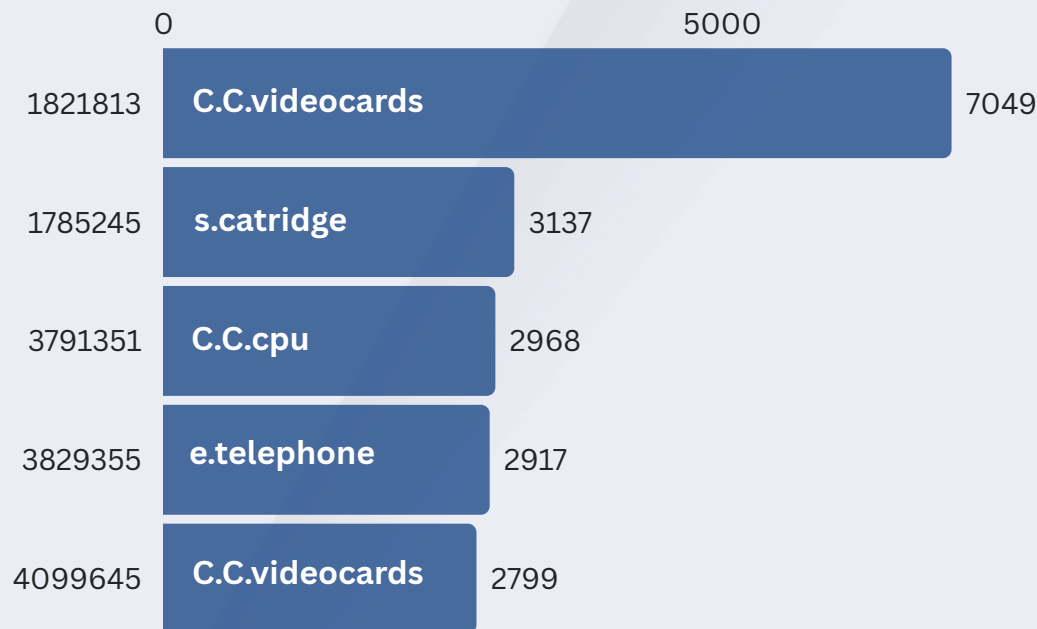


View-to-Purchase Conversion Is Dominated by Stationery.Cartridge and Electronics.Telephone Products

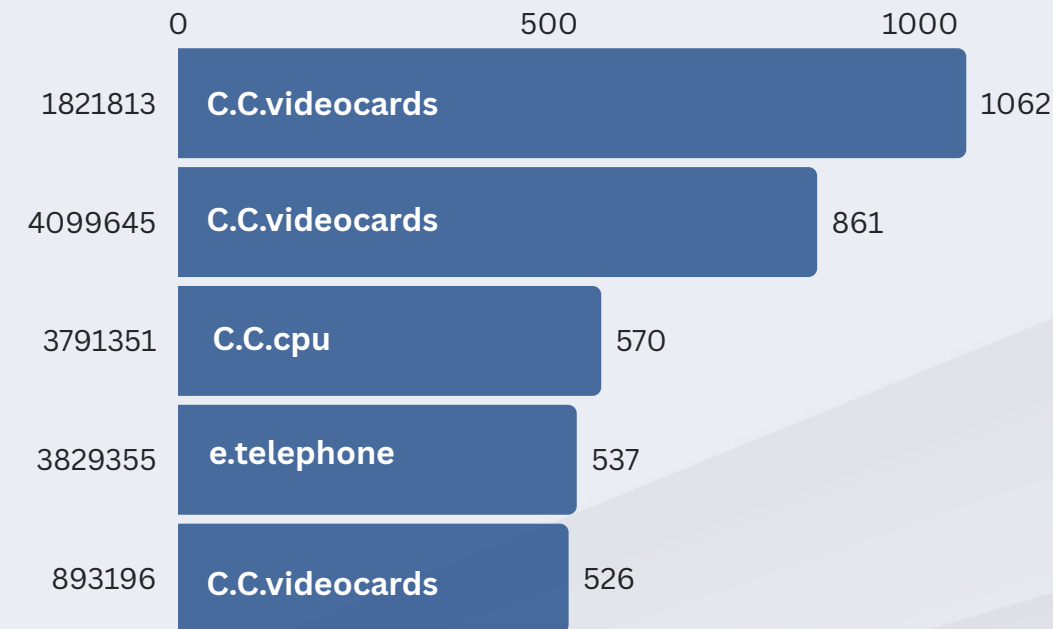
Top 5 Product IDs

The Highest View, Cart, and Purchase

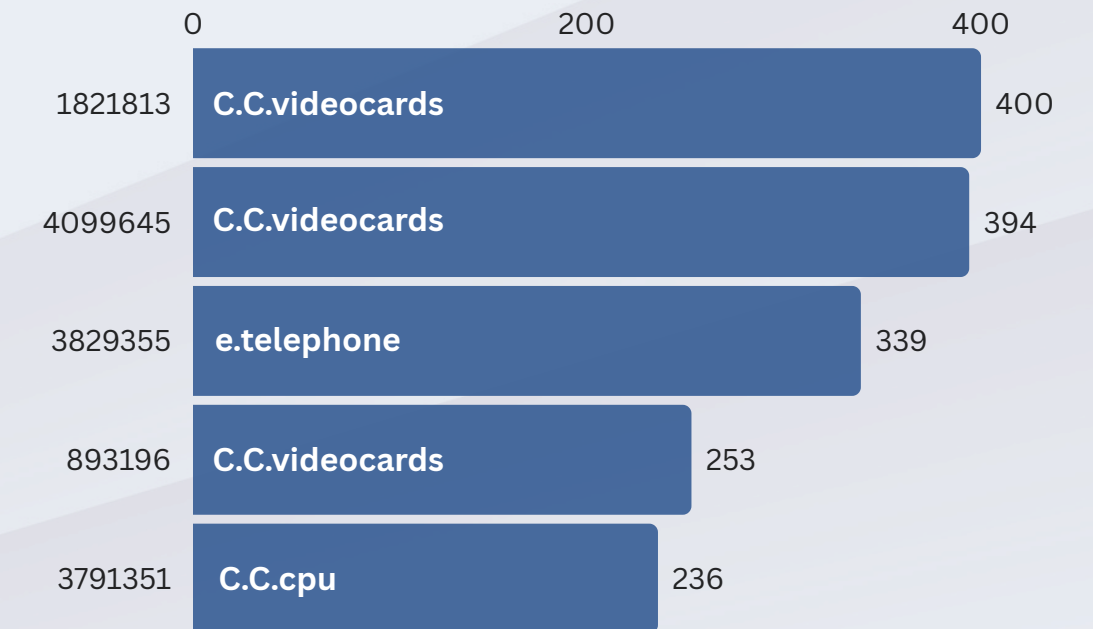
View



Cart



Purchase



The c.c. videocards, telephone, and CPU categories dominate customer engagement across the funnel, from views to purchases. Product ID 1821813 ranks first in views, cart additions, and purchases, demonstrating strong conversion performance and sustained customer interest

BUSINESS RECOMMENDATION

products performance

Category	Goal	Recommendation
Top 5 Viewed Products	Improve conversion	• Optimize pricing strategy • Add promotional banners • Use these products in retargeting ads • Highlight them as “Best Sellers” • Improve visibility of add-to-cart button
Top 5 Carted Products	Increase purchase completion	• Send cart abandonment emails • Offer small checkout incentives • Reduce checkout steps • Provide price transparency (no hidden fees)
Top 5 Purchased Products	Maximize revenue & leverage demand	• Bundle with related products • Increase inventory stock • Use as anchor products in marketing campaigns • Upsell premium versions • Feature them on homepage

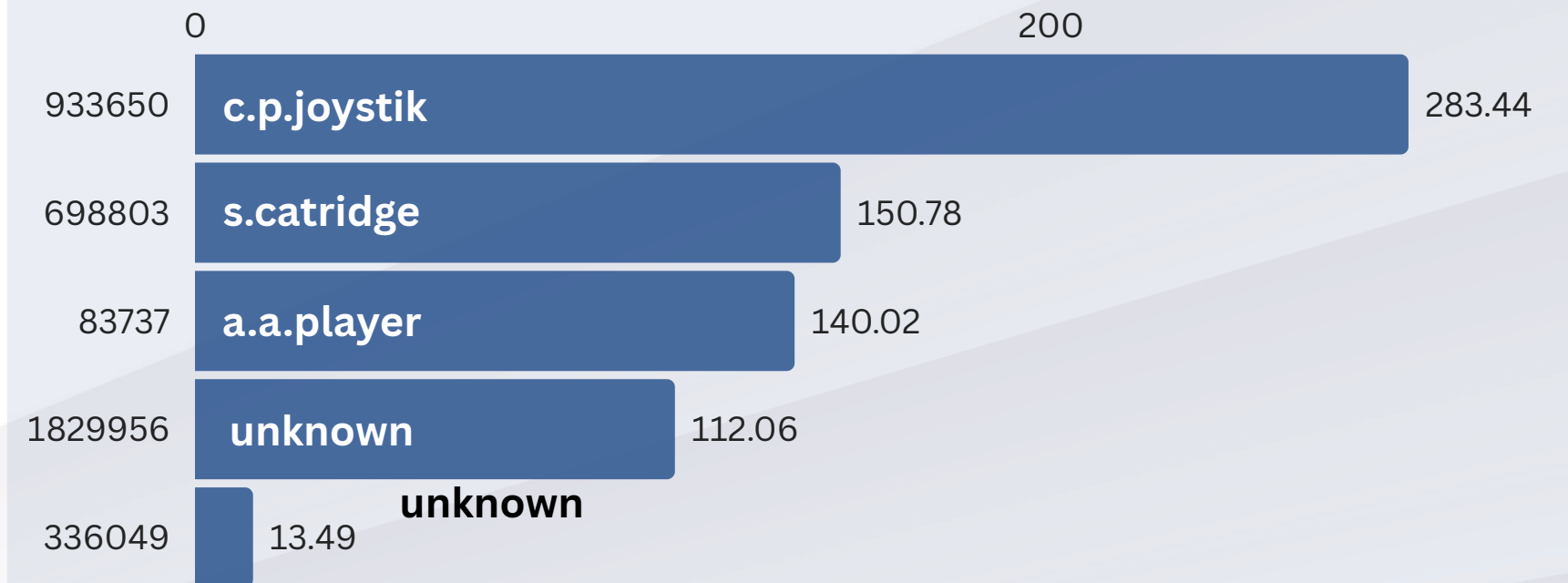
Top 5 Product IDs

100% drop view → Cart and cart → Purchase

drop view → cart



drop cart → purchase



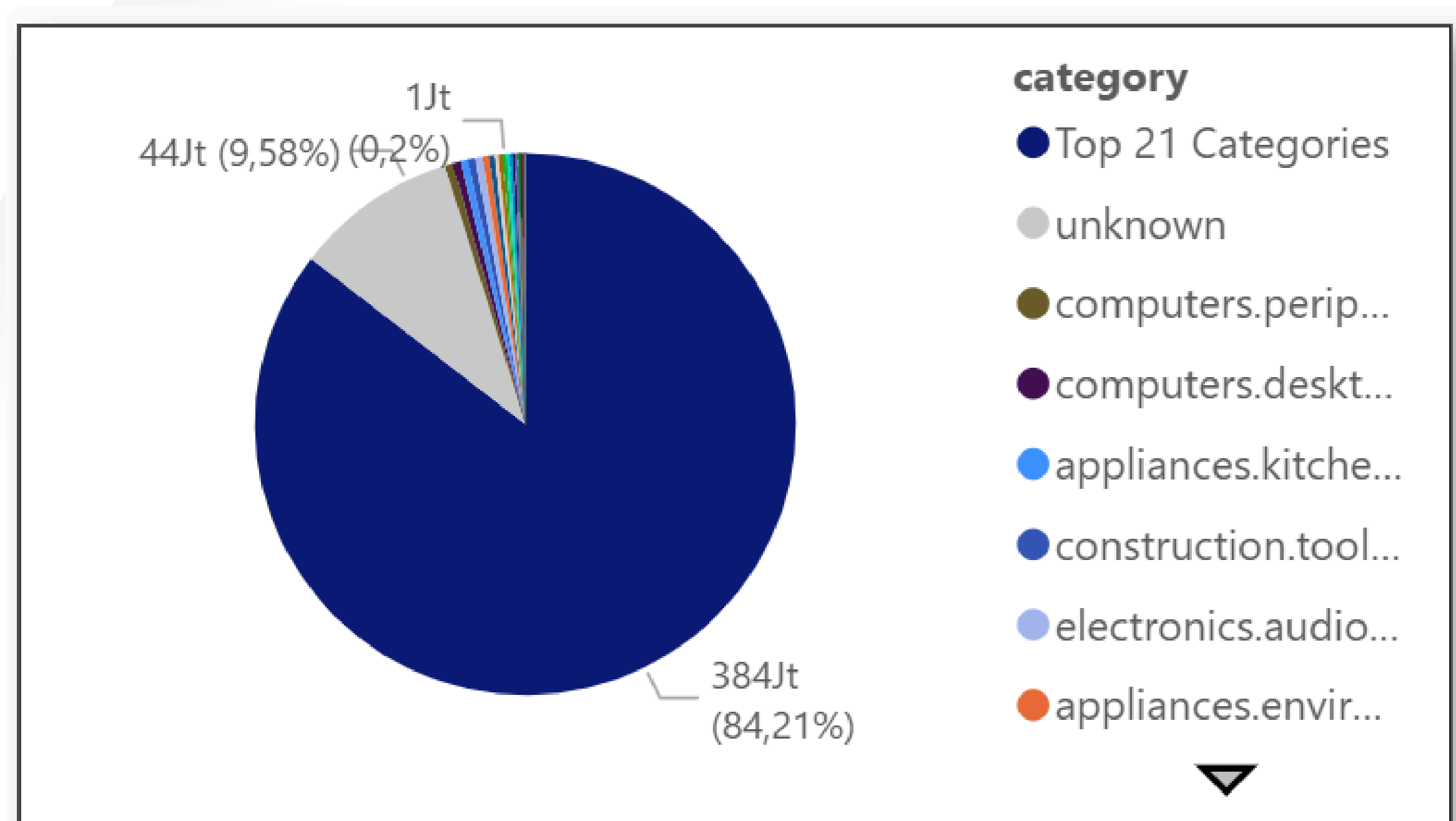
Unclear product categorization and higher prices contribute significantly to drop-offs, highlighting the importance of clear information and affordable pricing to improve conversion.

BUSINESS RECOMMENDATION

products performance

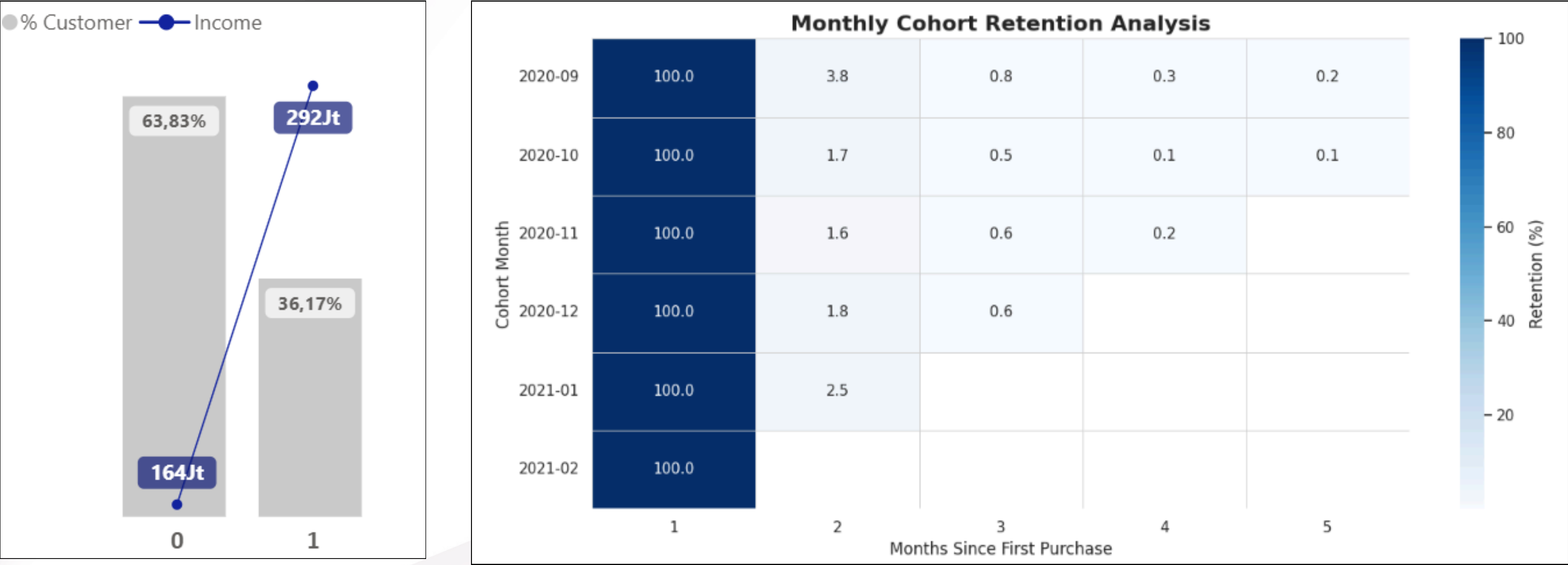
category	Insight	Goal	Recommendation
Drop View → Cart	Unclear categories and weak product clarity reduce add-to-cart behavior	Increase add-to-cart rate	• Improve product descriptions & categorization • Add high-quality images & detailed specs • Highlight product benefits clearly • Show customer reviews & ratings • Add urgency indicators (limited stock / discount badges)
Drop Cart → Purchase	High-priced products (> \$100) dominate drop-offs.	Reduce checkout friction	• Offer installment/payment plans • Provide limited-time discounts • Add trust badges (secure payment, warranty) • Offer free shipping thresholds • Simplify checkout process

20% of Categories Contribute 84% of Total Revenue



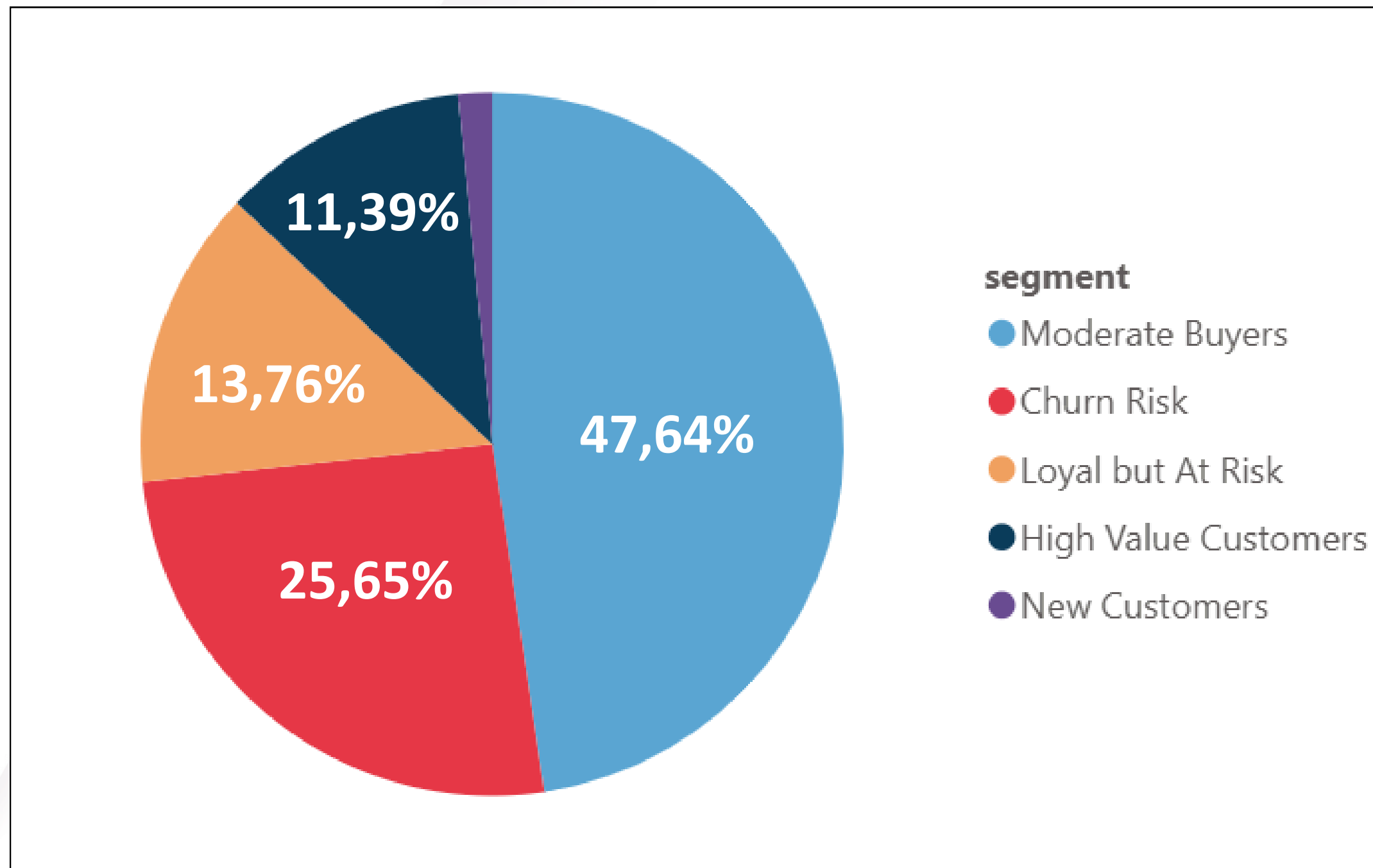
The top 21 categories (approximately 20% of total categories) contribute 84.21% of total revenue, indicating a strong Pareto distribution and high revenue concentration in a small number of categories

Only 36% of Customers Make Repeat Purchases



Repeat purchase rate remains low at 36%, and retention drops sharply after the first month. Although repeat customers generate higher revenue, long-term customer retention is still weak, indicating strong dependence on short-term purchases rather than sustained loyalty

Revenue Concentrated in High Value and Moderate Customers, While **25%** of Customers Are at Risk of Churn



Moderate Buyers

Total customer 10,15 rb

Churn Risk

Total customer 5,47 rb

Loyal but At Risk

Total customer 2,93 rb

High Value Customers

Total customer 2,43 rb

New Customers

Total customer 332

BUSINESS RECOMMENDATION

customer segmentation

Segement	Goal	Recommendation
High Value Customers	Increase retention & maximize lifetime value	• Offer exclusive loyalty rewards and early access to new products • Provide personalized product recommendations • Introduce premium bundles or cross-sell high-margin items • Implement VIP membership tiers
Moderate Buyers	Convert into High Value customers	• Targeted email campaigns with personalized promotions • Offer limited-time discounts to increase purchase frequency • Suggest complementary products (cross-selling) • Provide incentives for repeat purchases (e.g., loyalty points)
Loyal but At Risk	Prevent churn & re-engage	• Send reactivation campaigns with special discounts • Offer reminder emails for abandoned carts • Provide personalized product suggestions based on past purchases • Gather feedback to understand declining engagement

BUSINESS RECOMMENDATION

customer segmentation

Segement		Goal	Recommendation
Churn Customers	Risk	Recover & rebuild trust	• Launch win-back campaigns with strong incentives • Provide price discounts or free shipping • Improve communication and after-sales support • Highlight value proposition and product benefits
New Customers		Encourage second purchase	• Send post-purchase follow-up emails • Offer first-repeat purchase discount • Recommend related products • Provide onboarding content to build trust



Business Performance Overview

Selected time period : September 2020 - February 2021

Date

Semua

Month

Semua

Year

Semua

Overview

Product

Customer



455Jt

Total revenue

21,30Rb

Total customer

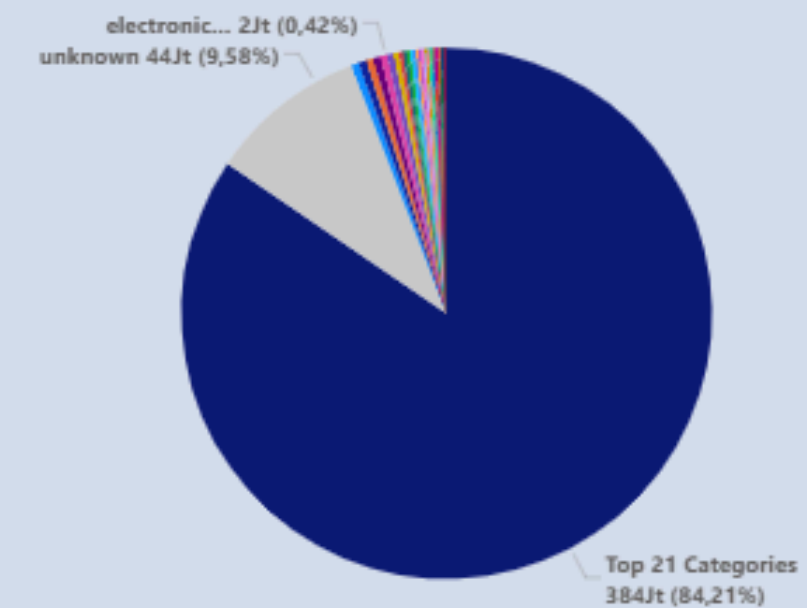
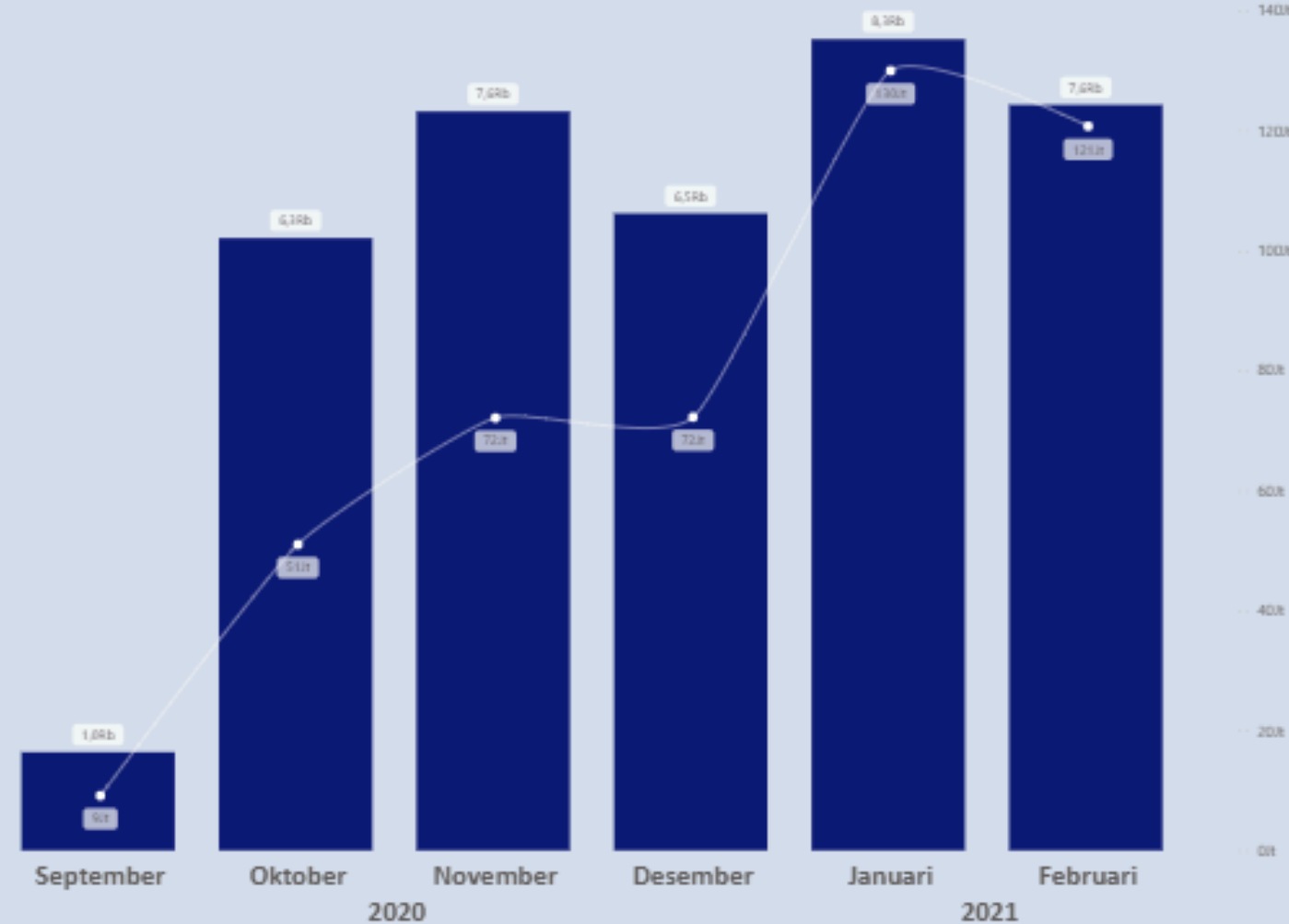
37,34Rb

Total order

36,17%

Repeat rate

● Total customer ● Total revenue





Product Performance

Selected time period : September 2020 - February 2021

Date

Semua

Month

Semua

Year

Semua

648Rb

Total View

43Rb

Total Cart

30Rb

Total Purchase

5%

Overall Conversion Rate %

Overview

Product

Customer

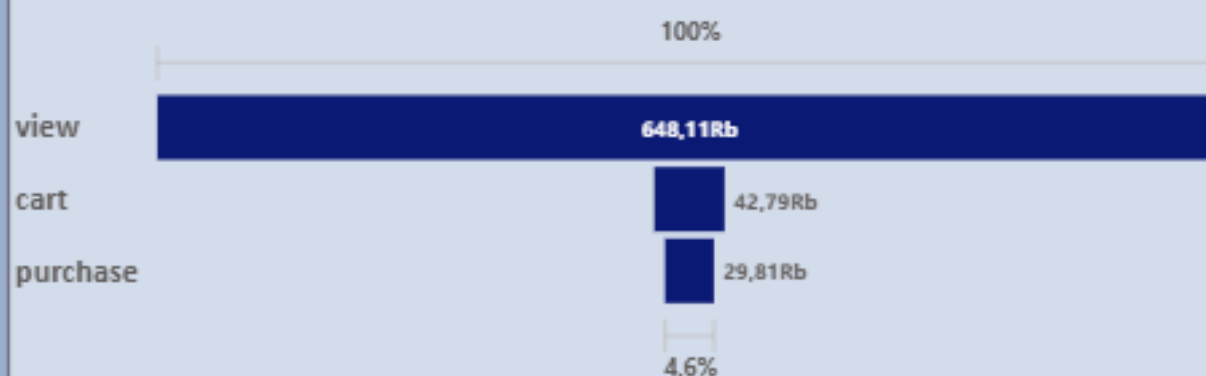
☐ cart

☐ purchase

☐ view

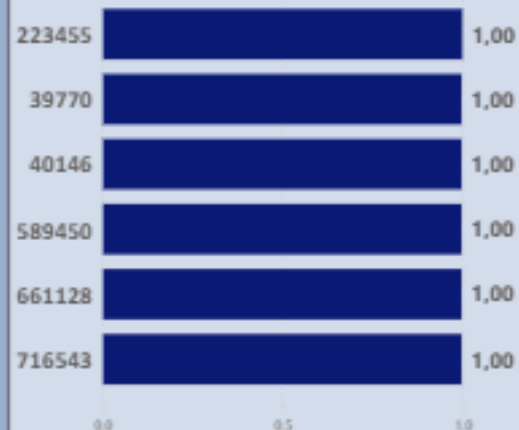


View to Purchase Funnel



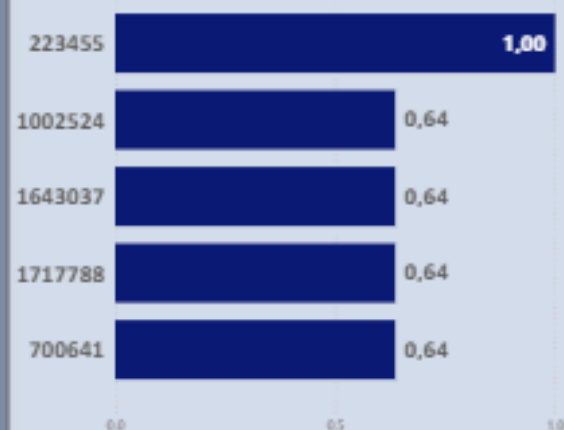
Top 5

View to Cart Drop off Rate (%)

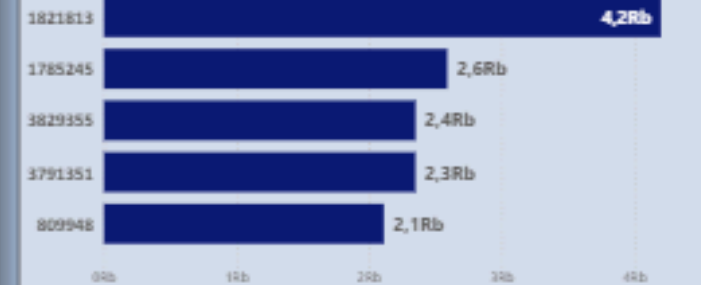


Top 5

Cart to Purchase Drop off Rate (%)



Top 5 view by customer



Top 5 cart by customer



Top 5 purchase by customer



Category

- ☐ Cari
- ☐ accessories.bag
 - ☐ appliances.environment.air_he...
 - ☐ appliances.environment.vacuum
 - ☐ appliances.iron
 - ☐ appliances.kitchen.blender
 - ☐ appliances.kitchen.coffee_grin...
 - ☐ appliances.kitchen.coffee_mac...
 - ☐ appliances.kitchen.fryer
 - ☐ appliances.kitchen.grill
 - ☐ appliances.kitchen.juicer
 - ☐ appliances.kitchen.kettle
 - ☐ appliances.kitchen.meat_grinder
 - ☐ appliances.kitchen.microwave
 - ☐ appliances.kitchen.mixer
 - ☐ appliances.kitchen.toster
 - ☐ appliances.personal.hair_cutter
 - ☐ appliances.personal.massager
 - ☐ appliances.personal.scales
 - ☐ appliances.sewing_machine
 - ☐ appliances.steam_cleaner
 - ☐ auto.accessories.alarm
 - ☐ auto.accessories.compressor
 - ☐ auto.accessories.gps
 - ☐ auto.accessories.immobilizer
 - ☐ auto.accessories.light
 - ☐ auto.accessories.parktronic
 - ☐ auto.accessories.player
 - ☐ auto.accessories.radar
 - ☐ auto.accessories.videoregister
 - ☐ auto.accessories.window
 - ☐ computers.components.cdrw
 - ☐ computers.components.cooler
 - ☐ computers.components.cpu



Customer behaviour

Selected time period : September 2020 - February 2021

Date

Semua

Month

Semua

Year

Semua

21,30Rb

Total customer

7706

Repeat Customer

36,17%

% Repeat Customer

64,04%

Repeat Revenue %

35,96%

New Revenue %

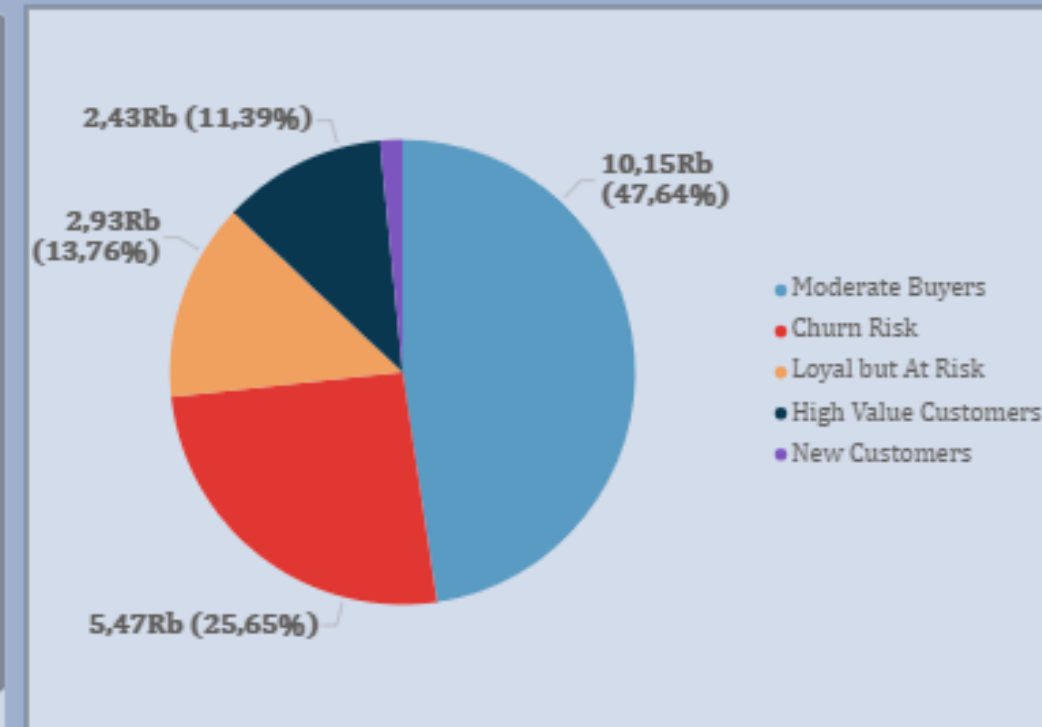
Overview

Product

Customer



Cohort Month	0	1	2	3	4
2020-09	100,00%	3,83%	0,80%	0,32%	0,16%
2020-10	100,00%	1,72%	0,45%	0,08%	0,11%
2020-11	100,00%	1,60%	0,61%	0,16%	
2020-12	100,00%	1,80%	0,56%		
2021-01	100,00%	2,50%			
2021-02	100,00%				



Total order by day



Total order by hour





THANK YOU

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